TEXORA · COURSE NOTES · LECTURE SHEET

Other Natural Fibres

Kapok, coir, the leaf fibres sisal and abaca, newer fruit and stem fibres, and the mineral fibre asbestos — the specialists that complete the natural-fibre map.

TE-1209 · NATURAL TEXTILE FIBRES · TOPIC 4 · LEVEL 1 TERM 2

1. Why these fibres form one topic

Topics 2 and 3 covered the high-volume natural fibres of clothing. The rest of the natural world's fibres are specialists: each does a small number of jobs better than any rival, usually because of one extreme property — kapok's buoyancy, coir's rot resistance, sisal's toughness in twine. They are classified by plant part like the cellulosics (most are cellulosic too), with one mineral exception at the end.

2. Kapok — the fibre that floats

Kapok is the seed hair of the kapok tree, a silky fibre surrounding the seeds in its pods. Its single cells are hollow tubes with air sealed inside and a waxy, water-repellent wall, making kapok extraordinarily light and buoyant — it can support many times its own weight in water — and a good thermal and sound insulator. The same waxiness and smoothness make it almost **unspinnable** into yarn: the fibres will not cohere. Kapok therefore works as a **filling** fibre — life jackets and buoyancy aids (its classic role), pillows, mattresses and insulation — an example of a fibre whose whole career is one outstanding property.

3. Coir — the fibre that survives the sea

Coir is extracted from the husk of the coconut — the only major **fruit fibre**. Husks are retted (often in brackish water for months) and beaten to release the fibres: long *bristle* fibres and shorter *mattress* fibres. Coir is heavily lignified, coarse, springy and exceptionally resistant to rot and salt water — among the most durable of all natural fibres. Its uses follow: mats and matting, brushes and brooms, ropes and fishing nets, mattress and upholstery padding, and coir geotextiles that protect soil and riverbanks while slowly biodegrading into them.

4. The leaf fibres: sisal and abaca

Leaf fibres are scraped or decorticated from the long leaves of monocot plants, where they run as stiffening strands.

- **Sisal** (from the agave plant) is strong, coarse and durable — the standard fibre for **agricultural twine**, ropes, sacks, mats and brushes, and a reinforcement in composites and plaster.
- **Abaca (Manila hemp)**, from a banana relative in the Philippines, is among the strongest of all natural fibres and famously resistant to salt-water damage; it made the world's best marine ropes (the name "Manila rope") and is still used for specialty papers — tea bags, banknotes and filters — where wet strength matters.

5. Newer and regional plant fibres

- **Pineapple leaf fibre (PALF)** — fine, white and strong for a leaf fibre; woven in the Philippines into the sheer piña cloth, and researched widely (including in Bangladesh) as a sustainable apparel and composite fibre.
- **Banana fibre** — separated from the banana pseudostem; lustrous and reasonably strong, used in mats, bags, handicrafts and blends, turning a plantation waste into a product.
- **Sunn hemp, mesta (kenaf)** — bast fibres related to jute, grown widely in South Asia for cordage, sacks and pulp; kenaf is also a composite and automotive-panel fibre.

The thread connecting them is the modern one: these are **agro-waste and low-input fibres**, and research effort is going into making their extraction and spinning economical, which is exactly the kind of work awaiting textile engineers in fibre-producing countries.

6. The mineral fibre: asbestos

Asbestos is the course classification's one natural **mineral** fibre — a group of silicate minerals that split into fine, flexible fibres, incombustible and chemically resistant, long used for fireproof clothing, insulation and brake linings. It is included to complete the classification and as a professional warning: inhaled asbestos fibres cause incurable lung diseases (asbestosis and mesothelioma), and its use is now banned or heavily restricted in most countries, replaced by glass, aramid and other man-made fibres. Fibre science includes knowing what not to use, and why.

7. How the specialists are extracted

The minor fibres share the bast fibres' problem — the useful fibre is locked inside plant tissue — but each has its own key that turns the lock. **Coir:** coconut husks are soaked (retted) in brackish pits or lagoons for months until the binding matter softens, then the husks are beaten on spikes or in defibering machines, separating the long bristle fibres from the short mattress fibres and the pith dust; the fibre is

then washed, dried and baled. **Leaf fibres** cannot be retted (the fibre would rot with the leaf), so sisal and abaca leaves are **decorticated** — crushed and scraped mechanically so the soft leaf pulp is stripped away from the fibre strands, which are then washed and sun-dried; the whole art is to scrape hard enough to clean the fibre without cutting it. **Kapok** needs no freeing at all: the pods are dried until they open, and the fluffy fibre is separated from the seeds by air blowing and hand picking — which is also why it stays unspun, arriving as loose floss. **Banana and pineapple fibres** are extracted by the leaf/stem logic — scraping (decortication), sometimes after a brief retting — and their quality depends heavily on how fresh the waste material is when processed.

8. All the specialists compared

FIBRE	ORIGIN	EXTREME PROPERTY	MAIN USES
Kapok	Seed hair	Buoyancy, lightness, insulation	Life jackets, stuffing, insulation
Coir	Coconut husk	Rot and salt-water resistance	Mats, brushes, ropes, geotextiles
Sisal	Leaf	Toughness at low cost	Twine, rope, mats, composites
Abaca	Leaf	Strength + marine durability	Marine rope, specialty papers
PALF	Leaf	Fineness among leaf fibres	Piña cloth, blends, composites
Banana	Pseudostem	Lustre; uses a waste stream	Mats, bags, handicrafts
Asbestos	Mineral	Fire resistance — but carcinogenic	Historical only; replaced by man-made fibres

9. Glossary — key terms of this topic

TERM	MEANING
Decortication	Mechanical scraping or crushing to free leaf or stem fibres from the surrounding tissue.
Fruit fibre	A fibre obtained from the husk or fruit wall of a plant — coir is the standard example.
Leaf fibre	A fibre extracted from leaves, such as sisal, abaca and pineapple leaf fibre.
Geotextile	A fabric laid on soil for reinforcement, drainage or erosion control.

10. Key points and summary

- The minor natural fibres are specialists: kapok floats, coir resists rot, sisal binds, abaca endures the sea — each career rests on one extreme property.
- Leaf and fruit fibres extend the cellulosic family beyond seed and bast; pineapple and banana fibres show the modern agro-waste direction of fibre research.
- Asbestos completes the classification as the mineral fibre, and its history is the profession's standing lesson in fibre safety.
- TE-1209's map is now complete: classification and essential properties (Topic 1), cellulosic fibres (Topic 2), protein fibres (Topic 3), and the specialists (Topic 4).

11. Practice

Answer in your own words; brief model answers follow.

1. Kapok is described as "almost unspinnable". Give the two structural reasons, and state the product form in which kapok is therefore used.
2. A coastal engineer needs a rope that will spend years in sea water, and a mat for a wet jetty. Choose fibres from this topic for each and justify.
3. Why are leaf fibres decorticated rather than retted? What can go wrong in decortication?
4. Abaca is used for banknote and tea-bag papers. Connect this use to two of its fibre properties.
5. A jute-mill region has banana plantations nearby. Propose one fibre product the region could develop, naming the waste stream it uses and one technical problem that must be solved first.
6. State the classification position of asbestos and the professional lesson its history teaches.

Answer key

1. Its fibres are smooth and waxy with no cohesion, and hollow/brittle — they cannot grip in a yarn. Used as loose filling (floss).
2. Rope: abaca (marine rope tradition, salt-water resistance) — coir also acceptable. Mat: coir — springy, rot-resistant, made for wet floors and jetties.
3. Retting would rot the leaf fibre along with the pulp; decortication strips the pulp mechanically. Over-scraping cuts and weakens the fibres; under-scraping leaves pulp that rots and stains.
4. Wet strength (the sheet must survive boiling water in a tea bag and handling as currency) and durability/folding endurance from long, strong fibres.
5. Banana fibre products (mats, bags, blended fabrics) from pseudostem waste; the key problem is economical, uniform extraction — fibre quality varies with freshness and the scraping process, and spinning-grade consistency must be achieved.

6. A natural mineral fibre (the classification’s inorganic branch). Lesson: a fibre’s technical excellence never overrides its health safety — performance is judged together with the hazard to makers and users.
